Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/27/2008
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Countermeasures for Space Radiation Induced Myeloid Leukemia		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	akennedy@mail.med.upenn.edu	**Fax:**	FY 215-898-0090
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-0079
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	Oncology Research Division		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	6	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gewirtz, Alan (University of Pennsylvania) Biaglow, John (University of Pennsylvania) Ware, Jeffrey (University of Pennsylvania) Stewart, Jelena (University of Pennsylvania) Newberne, Paul (Boston Medical Center)		
Grant/Contract No.:	NCC 9-58-IIH00402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposed research investigated mechanisms to resist malignancy in space with an emphasis on evaluation of certain nutritional supplements as countermeasures to protect against space radiation-induced leukemogenesis; these nutritional supplements include L-selenomethionine (SeM), vitamin C, vitamin E succinate, alpha-lipoic acid, Co-enzyme Q10 and N-acetylcysteine. Data obtained from our previous in vitro and animal studies supported by the National Space Biomedical Research Institute (NSBRI) have demonstrated that the nutritional supplements evaluated in this research project are effective in preventing radiation induced oxidative stress and protecting cultured human cells against adverse biological effects induced by the types of radiation of most concern during space travel, i.e., highly energetic heavy charged particles (known as HZE particles) and protons. This investigation determined the efficacy of nutritional supplements as countermeasures against space radiation induced leukemogenesis using the CBA mouse model system. Two 2-year animal experiments were performed to determine the effect of the supplement combination on the development of leukemogenesis in CBA mice exposed to radiation with HZE particles (1-GeV/nucleon iron ions) or protons. In addition, several short-term animal and cell culture experiments were performed to determine the effects of radiation treatment and nutritional supplementation on selected surrogate endpoint biomarkers (SEBs), which are the host bio-reduction capacity measured as plasma total antioxidant status and OXR1 gene expression in selected populations of white blood cells. The results of this study have provided critical information about the feasibility and mechanism(s) of nutritional supplements in increasing the resistance to space radiation induced malignancy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	People living on earth are exposed to higher than normal doses of radiation from a variety of sources and for a variety of reasons. It is expected that the countermeasures being developed as part of this NSBRI research project will be equally useful for the protection of people on earth from the expected adverse effects of exposure to ionizing radiation.
Task Progress:	This research program involved studies of nutritional supplements as countermeasures to protect against the development of space radiation-induced malignancies. In this study, two dietary supplement preparations were evaluated for their ability to prevent the development of malignancies in CBA mice exposed to different forms of space radiation: protons and highly energetic, heavy particles (HZE particles). One preparation contained a mixture of antioxidant agents (L-selenomethionine [SeM], vitamin C, vitamin E succinate, alpha-lipoic acid, coenzyme Q10 and N-acetyl cysteine). The other contained the soybean-derived Bowman-Birk protease inhibitor (BBI), utilized in the form of BBI Concentrate (BBIC). The major finding was that there was a reduced risk of developing malignant lymphoma in animals exposed to space radiation and maintained on diets containing the antioxidant formulation or BBIC compared to the irradiated animals maintained on the control diet. In addition, the two different dietary countermeasures also reduced the yields of a variety of different rare tumor types (e.g., Harderian gland tumors) observed in the animals exposed to space radiation. The results suggest that dietary supplements could be useful in the prevention of malignancies and other neoplastic lesions developing from exposure to space radiation. Studies were also performed to determine whether a dietary supplement consisting of SeM, vitamin C, vitamin E succinate, alpha-lipoic acid and N-acetyl cysteine could improve the survival of mice following total body irradiation. Antioxidants significantly increased the 30-day survival of mice following exposure to a potentially lethal dose of X-rays when given prior to or after animal irradiation. Pretreatment of animals with antioxidants resulted in significantly higher total white blood cell and neutrophil counts in peripheral blood at 4 and 24 hours following 1 Gy and 8 Gy. Antioxidants were effective in preventing peripheral lymphopenia only after low dose irradiation. Antioxidant supplementation was also associated with increased bone marrow cell counts following irradiation. Supplementation with antioxidants was associated with increased Bcl-2, and decreased Bax, caspase-9 and TGF-alpha1 mRNA expression in the bone marrow following irradiation. Maintenance of the antioxidant diet was associated with improved recovery of the bone marrow following sub-lethal or potentially lethal irradiation. Taken together, oral supplementation with antioxidants appears to be an effective approach for radioprotection of hematopoietic cells, improvement of animal survival, and modulation of apoptosis is implicated as a mechanism for the radioprotection of the hematopoietic system by antioxidants.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Arbogast S, Smith J, Matuszczak Y, Hardin BJ, Moylan JS, Smith JD, Ware J, Kennedy AR, Reid MB. "Bowman-Birk inhibitor concentrate prevents atrophy, weakness, and oxidative stress in soleus muscle of hindlimb-unloaded mice." J Appl Physiol. 2007 Mar;102(3):956-64. PMID: 17110517 , Mar-2007
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Articles in Peer-reviewed Journals	Kennedy AR, Ware JH, Guan J, Donahue JJ, Biaglow JE, Zhou Z, Stewart J, Vazquez M, Wan XS. "Selenomethionine protects against adverse biological effects induced by space radiation." Free Radic Biol Med. 2004 Jan 15;36(2):259-66. PMID: 14744637 , Jan-2004
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